BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 131 · Lesson 131 · Asymmetric Encryption and Public-Key Use

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the principal challenge of symmetric encryption between distant users?

Q2. What is a function parameter?

Q3. What is the binary result of 1 + 1 in a full adder with carry-in 0?

Q4. How many bits are in three bytes?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

public key — A key that can be distributed openly.

private key — A matching secret key kept by its owner.

asymmetric encryption — Uses a related public/private key pair.

digital signature — Cryptographic evidence of origin and integrity with private-key signing.

### New learning

Generate key pair — Public key can be shared; private key remains secret.

Encrypt to recipient — A message encrypted with recipient public key is decrypted by recipient private key.

Sign and verify — Signature created with private key is verified using corresponding public key.

### Worked example

Alice wants to send Bob a private message using public-key encryption. Identify both keys used.

1. Bob shares public key — Predict the first step from the prompt.

2. Alice encrypts for Bob — Explain the intermediate result or decision.

3. Bob uses private key — Alice encrypts a suitably sized message or session key using Bob's public key. Bob decrypts using his matching private key. Alice never requires Bob's private key.

# We Do · Practise with guidance

A website publishes a signed update. Which key verifies its authentic signature?

Publisher signs privately

Predict the first step from the prompt.

Distribute signed file

Explain the intermediate result or decision.

Verify with public key

The publisher creates the digital signature with its private key. Users verify it with the public key obtained through a trusted route; this supports authenticity and integrity, not secrecy of the update.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain the difference between public and private keys and how they support confidentiality. [6]

### Question 2

Outline how a digitally signed download is verified. [6]

### Question 3

Compare asymmetric and symmetric approaches for sending a large encrypted video file. [8]

## Exit Ticket · Answer before checking

A public key must be kept secret for public-key encryption to work.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Securely sharing and managing the secret key.

Q2. An input used by a subroutine.

Q3. Sum 0, carry-out 1.

Q4. 24.

### You Do pages · Independent questions

Q1. Public key distributed [2], private key secret [2], public encrypt/private decrypt for recipient [2].

Q2. Signer signs using private key [2], verifier uses trusted public key [2], checks authenticity/integrity [2].

Q3. Symmetric efficient for bulk [2], asymmetric useful for secure session-key exchange [2], hybrid approach [2], key identity/trust and limitations [2].

### Exit Ticket · Model answer

False. The matching private key must be kept secret.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**